Requirements Definition

Cross-Border Bill Payment Platform
Canada → Japan | FinTech Project

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Table of Contents

1. Introduction 4

1.1 Purpose of This Document 4

1.2 Background 4

1.3 Reference Materials 4

1.4 Project Positioning and Assumptions 4

2. System Overview 5

Actors 7

External Systems and Institutions 7

Reference: Mapping to the MyBeacon Model 8

3. Conceptual Model 9

4. Functional Overview 9

5. Functional Requirements 10

5.1 Authentication 10

5.2 Bill Registration and Management 10

5.3 Payment Request 10

5.4 Exchange Rate and Fee Estimate 10

5.5 Transfer Integration (Wise) 11

5.6 Payment Status Tracking and Transaction History 11

5.7 Notifications 11

5.8 Admin Dashboard (Payment Agency) 11

5.9 Incoming Transfer Reconciliation 11

5.10 Payment Request Validation 12

5.11 Pay-easy Processing Queue 12

5.12 Manual Payment Confirmation 12

5.13 Auditing and Reporting 12

5.14 Payment Status Management 12

6. Non-Functional Requirements 14

6.1 Security 14

6.2 Compliance 14

6.3 Reliability and Consistency 14

6.4 Performance 14

6.5 Responsive Design 14

6.6 Maintainability 14
6.7 Scalability 15
6.8 Availability 15
7. Supported Environments 15
8. Development Roadmap 16
Phase 1 — Proof of Concept 16
Phase 2 — Payment Integration 16
Phase 3 — Operational Automation 16
9. Key Technical Challenges 16
10. Budget and Infrastructure 17
11. Project Goal 17
12. Disclaimer 17

1. Introduction

1.1 Purpose of This Document

This document defines the requirements for the Cross-Border Bill Payment Platform (hereinafter "the System"), which enables residents of Canada to pay Japanese bills — such as utilities, taxes, and insurance premiums — through a single application.

This document is not intended to contain only finalized information from the outset. It records what is known at present and will be refined step by step through alignment with stakeholders. Once complete, it will serve as the basis for the basic design and detailed design documents.

To avoid fragmentation of materials, any document related to the System will be linked from this document so that all relevant information can be traced from here.

1.2 Background

Some residents of Canada continue to have financial obligations in Japan, including electricity, water and gas, telecommunications and internet, national and local taxes, and insurance premiums. These payments often require a Japanese bank account or domestic payment methods, which makes paying from abroad cumbersome.

Meanwhile, MyBeacon, a Canadian FinTech company, provides an India Bill Pay service that allows customers in Canada to pay bills in India through partnerships with YES BANK and Bharat Connect (BBPS). This model shows that a FinTech application can access a large biller network without integrating directly with every individual biller.

The System adapts this model to Japan. By combining Wise (international transfers and currency conversion), a Japanese payment agency, and Pay-easy (Japan's multi-payment network), it aims to let customers pay their Japanese bills from Canada in one place.

1.3 Reference Materials

No.	Document	Summary
1	Cross-Border Bill Payment Platform README (GitHub)	Concept, architecture, and roadmap of the System
2	MyBeacon vs. Japan payment flow comparison (diagram)	Comparison of MyBeacon (Canada → India) and the System (Canada → Japan)
3	Wise Business / Wise Platform official documentation	To confirm international transfer and API integration eligibility (not yet investigated)
4	Pay-easy (Japan Multi-Payment Network Promotion Council) official information	To confirm supported billers and terms of use (not yet investigated)

1.4 Project Positioning and Assumptions

- This project is at the **Concept / Proof of Concept (PoC) stage**.
- The integrations with Wise, the payment agency, and Pay-easy described in this document are **proposed components**. They do not imply that commercial partnerships, regulatory approvals, or production payment integrations have been established.
- Direct API access to Pay-easy is not assumed.
- **No real customer funds are handled** in the initial PoC.

2. System Overview

The System converts CAD to JPY and transfers it through Wise for Japanese bills registered by customers in Canada. A Japanese payment agency then pays each biller using Pay-easy.

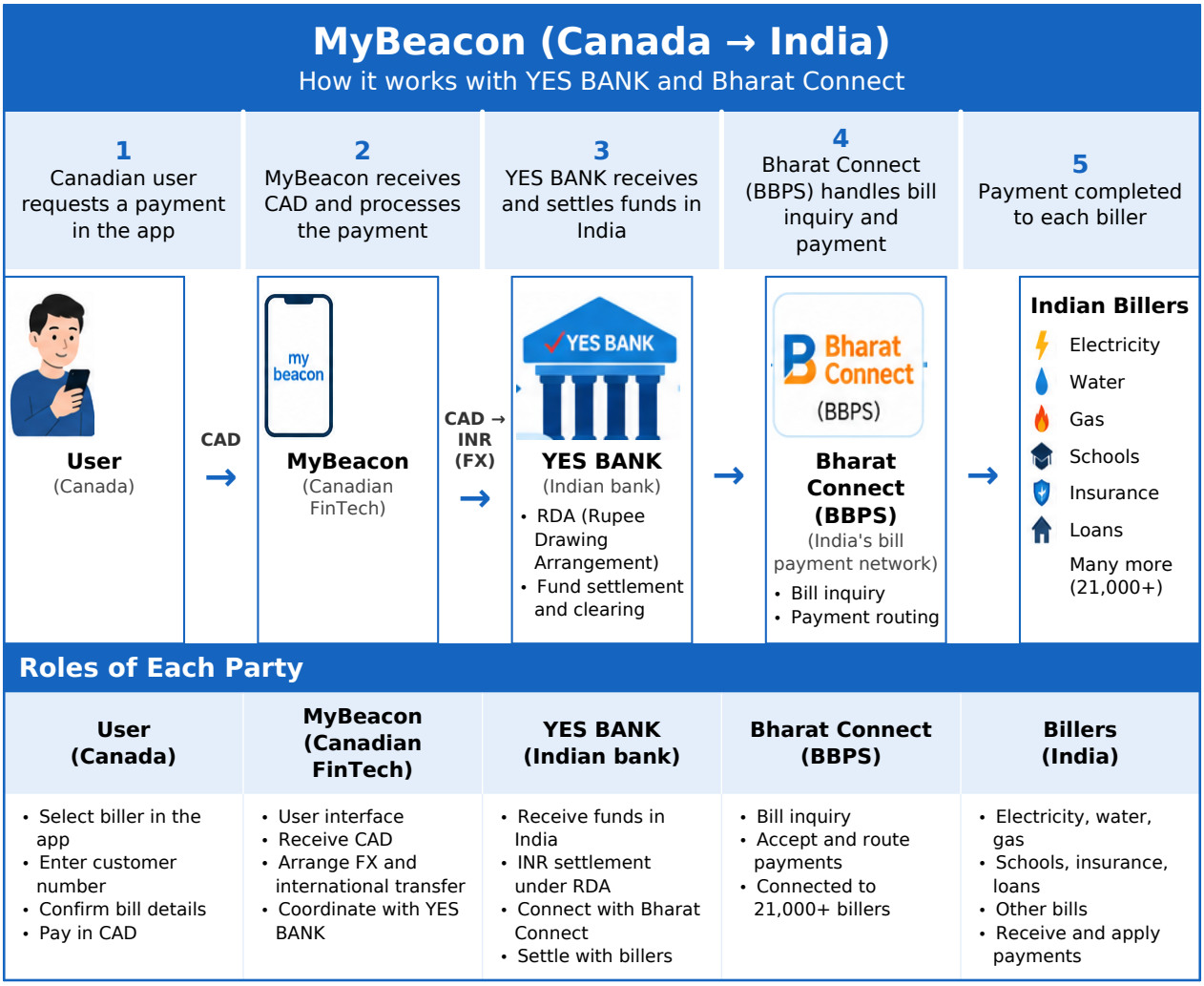


Figure 2-1 Reference model: MyBeacon India Bill Pay (Canada → India)

This Project (Canada → Japan)

App → Wise → Japanese Payment Agency → Pay-easy

1	2	3	4	5
Canadian user requests a payment in the app	JPY is sent to the Japanese corporate account via Wise	The Japanese payment agency receives the funds and pays via Pay-easy	Payment completed to each biller via Pay-easy	Payment completed to each biller



User
(Canada)

CAD →

Wise
(Intl. transfer & FX)

- CAD → JPY conversion
- Transfer to the Japanese corporate bank account

JPY →

Japanese Payment Agency

- Receive JPY in corporate account
- Manage payments per user
- Pay via Pay-easy from corporate internet banking
- Record payment results in the system
- Notify users

→

Pay-easy
(Japan Multi-Payment Network)

- Bill inquiry
- Payment processing
- Connected to billers nationwide

→

Japanese Billers

- Electricity
- Water
- Gas
- Telecom
- Taxes (national / local)
- Insurance
- Mail-order payments
- Many more

Roles of Each Party

User (Canada)	Wise (Intl. transfer & FX)	Japanese Payment Agency	Pay-easy (Japan Multi-Payment Network)	Billers (Japan)
• Register biller information in the app • Enter required details (biller number, customer number, etc.) • Confirm payment amount • Pay in CAD	• CAD → JPY conversion • Transfer to the Japanese corporate bank account • Incoming payment notification and identification	• Receive JPY in corporate account • Manage payments per user • Pay via Pay-easy from corporate internet banking • Record and notify payment results • Customer support	• Bill inquiry • Payment processing • Connected to billers nationwide	• Electricity, water, gas • Telecom, taxes • Insurance, mail-order payments • Other bills • Receive and apply payments

Figure 2-2 This project: Cross-Border Bill Payment Platform (Canada → Japan)

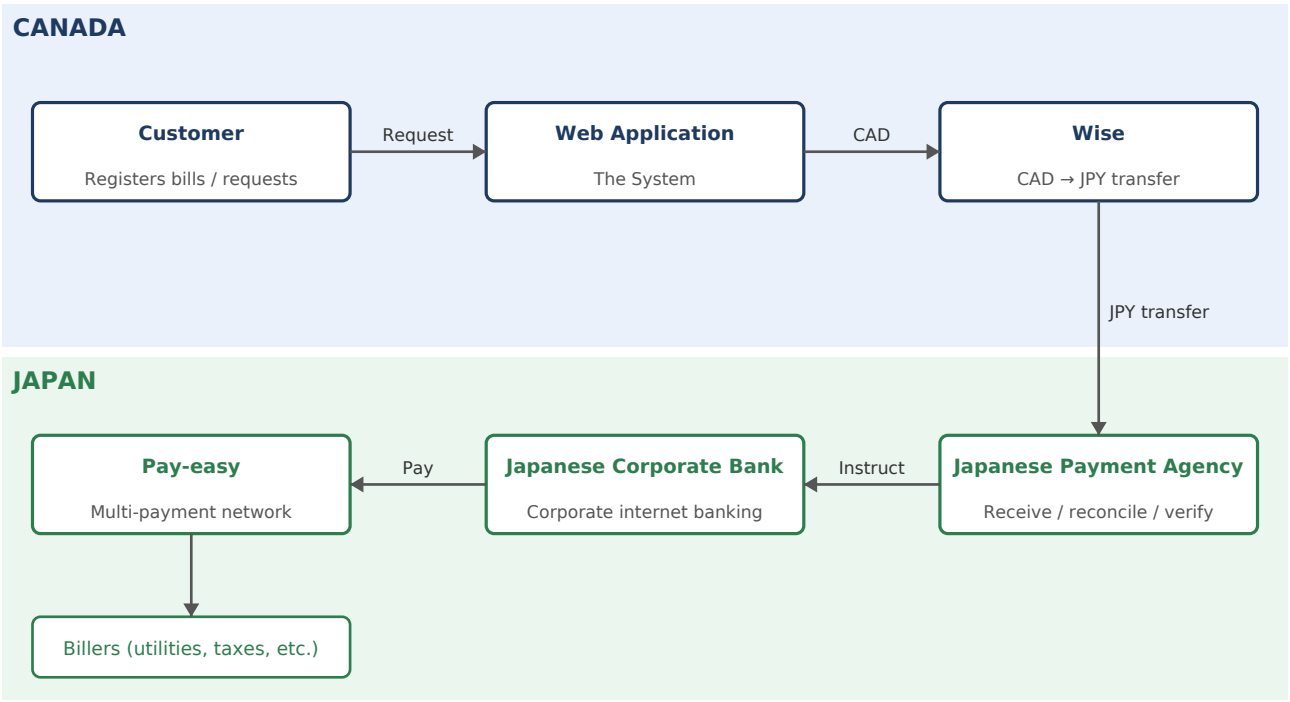


Figure 2-3 Overall configuration of the System (high-level architecture)

* Wise integration, payment agency operations, and Pay-easy payment execution are proposed components. Direct API access to Pay-easy is not assumed.

Actors

Name	Description
Customer	An individual living in Canada who needs to pay Japanese bills. Registers bills, submits payment requests, and checks payment status in the application.
Japanese Payment Agency	A legal entity established in Japan. Receives customer transfers in JPY, reconciles incoming funds, verifies payment requests, and pays domestic bills on behalf of customers.
Agency Operator	Staff member of the payment agency authorized to confirm incoming funds, process payments, and record results in the admin console.
System Administrator	Operates and maintains the System. Manages operator accounts and settings.
Billers	Organizations that receive payments and apply them to customer obligations, such as electric power companies, water bureaus, gas companies, telecom carriers, national/local governments, and insurers.

External Systems and Institutions

Name	Role
Wise	Handles CAD-to-JPY conversion and international transfers. Account type and API integration eligibility to be confirmed.
Japanese Corporate Bank	Provides the payment agency's corporate account and Pay-easy-enabled corporate internet banking.
Pay-easy	Japanese payment infrastructure that processes supported tax, utility, and other payments through participating financial institutions and billers.
Notification service (e.g., email)	Notifies customers of changes in payment status. Specific service to be determined.

Reference: Mapping to the MyBeacon Model

MyBeacon (Canada → India)	The System (Canada → Japan)	Role
Customer (Canada)	Customer (Canada)	Submits payment requests in the app and pays in CAD
MyBeacon	Web Application	Customer-facing interface; coordinates payments
YES BANK (RDA)	Wise + Japanese Payment Agency	Currency conversion and international transfer; local fund receipt and settlement
Bharat Connect (BBPS)	Corporate Bank + Pay-easy	Bill information inquiry; payment routing to billers
Indian Billers	Japanese Billers	Receive and apply payments

3. Conceptual Model

The main concepts (entities) handled by the System and their relationships are shown below.

Concept	Description	Key attributes (draft)
Customer	A Canadian resident registered with the System	Name, email address, identity verification (KYC) status
Bill (registered bill)	A Japanese bill registered by a customer. A customer can register multiple bills.	Biller, customer/contract number, payment reference number, amount, due date, Pay-easy identifiers (biller number, customer number, confirmation number)
Payment Request	A request by a customer to pay a registered bill. Multiple requests may occur for one bill.	Request number, target bill, JPY amount, estimated CAD amount, fees, exchange rate, status
Transfer (incoming funds)	Funds received via Wise in the agency's corporate account	Received date/time, amount (JPY), transfer reference, reconciliation status
Payment Execution	A domestic payment made by an operator via Pay-easy	Execution date/time, executed by, result, confirmation details such as receipt number
Audit Log	Record of all state changes and operations	Operator, operation, before/after values, date/time

Relationships: Customer 1 – * Bill 1 – * Payment Request 1 – 0..1 Transfer (reconciled) / Payment Request 1 – 0..* Payment Execution (including retries).

* "Funds received" does not mean "the Japanese bill has been paid." The two are managed as separate concepts with separate statuses.

4. Functional Overview

An overview of the functions to be implemented in the System.

Category	Functions
Customer functions	Authentication / Bill registration and management / Payment request / Exchange rate and fee estimate / Transfer integration / Payment status tracking / Transaction history / Notifications
Payment agency functions	Admin dashboard / Customer and bill management / Incoming transfer reconciliation / Payment request validation / Pay-easy processing queue / Manual payment confirmation / Auditing and reporting
Common functions	Payment status management / Duplicate payment prevention / Audit logging

5. Functional Requirements

The requirements for the functions to be implemented in the System.

5.1 Authentication

- Customers can sign up with an email address and password.
- An email address verification message is sent upon sign-up.
- Customers can log in and log out.
- Passwords can be reset.
- Agency operators and system administrators log in through an admin console separate from the customer interface.
- Operators can be assigned roles (view only / reconciliation / payment execution / administrator).
- In production, only customers who have completed identity verification (KYC) can submit payment requests. (KYC is simulated in the PoC.)

5.2 Bill Registration and Management

- Customers can register, list, edit, and delete Japanese bills.
- Multiple bills can be registered.
- Depending on the bill type, the following information can be entered:
  - Biller (service provider)
  - Customer number / contract number
  - Payment reference number
  - Payment amount (JPY)
  - Due date
  - Pay-easy identifiers (biller number, customer number, confirmation number) — where applicable
- Customers can select the bill category (electricity, water/gas, telecom/internet, national/local tax, insurance premium, other).
- Missing required fields or invalid formats (number of digits, digits only, etc.) result in an error at registration.
- Bills with an approaching due date are highlighted in the list.
- Bills with a payment request in progress cannot be deleted.

5.3 Payment Request

- Customers can select a registered bill and create a payment request.
- Before the request is created, the estimate from 5.4 (CAD amount, fees, exchange rate) is shown on a confirmation screen.
- When the customer agrees on the confirmation screen, the request is confirmed and its status moves CREATED → PENDING_TRANSFER.
- A unique request number (transfer reference) is issued for each request and used to identify the transfer.
- A new request cannot be created for a bill that already has a request in progress (duplicate payment prevention).
- Customers can cancel a request before funds are received (PENDING_TRANSFER) (CANCELLED).

5.4 Exchange Rate and Fee Estimate

- The estimated payment amount in CAD is displayed for the JPY bill amount.
- The display includes the exchange rate, international transfer fee, service fee, and total CAD amount.
- A notice states that the estimate is approximate and may differ from the actual rate.
- The exchange rate source, validity period, and fee structure are to be determined (decided in Phase 2). The PoC uses fixed or simulated rates.

5.5 Transfer Integration (Wise)

- Customers send CAD using Wise, which arrives as JPY in the agency's corporate account.
- The transfer screen shows the destination account details and the request number (transfer reference).
- In the PoC, transfers are simulated and no real funds are moved.
- In Phase 2, the eligibility of Wise Business / Wise API and its fit with the customer-payment-to-Japanese-corporate-account model will be evaluated.

5.6 Payment Status Tracking and Transaction History

- Customers can view the status of each payment request (see 5.14).
- After completion, available confirmation details such as payment date/time and receipt number are displayed.
- Transaction history can be listed and filtered by period, bill, and status.
- Transaction history can be downloaded as CSV. (Low priority)

5.7 Notifications

- Notifications (email) are sent to customers at the following times:
  - When a payment request is created
  - When funds are received (FUNDS_RECEIVED)
  - When payment is completed (COMPLETED)
  - When payment fails (FAILED) or is cancelled (CANCELLED)
  - When an unpaid bill is approaching its due date (reminder)
- Notification channels and wording are to be determined. The PoC assumes email notifications only.

5.8 Admin Dashboard (Payment Agency)

- Displays the number of payment requests by status (awaiting funds, funds received, ready for payment, processing, failed, etc.).
- Displays items requiring attention, such as requests nearing their due date or requests with no incoming funds for an extended period.
- Customers and bills can be searched, listed, and viewed in detail.

5.9 Incoming Transfer Reconciliation

- Operators can register incoming fund information for the corporate account. (Manual entry or simulated data in the PoC)
- Incoming fund information (amount, transfer reference, date/time) can be matched and linked (reconciled) to a payment request.
- When the transfer reference and amount match, the request is automatically suggested as a reconciliation candidate.

- Mismatches such as over/under payment or unknown references are listed as exceptions and can be handled manually.
- When reconciliation is complete, the payment request status is updated to FUNDS_RECEIVED.

5.10 Payment Request Validation

- Operators can review the details of a payment request (biller, customer number, amount, due date).
- Operators confirm that sufficient funds are available for the payment amount.
- When validation is complete, the status is updated to READY_FOR_PAYMENT.
- If there are deficiencies, a reason for return can be recorded and the customer notified.

5.11 Pay-easy Processing Queue

- Payment requests in READY_FOR_PAYMENT are shown as a processing queue, ordered by nearest due date.
- Operators select an item from the queue and change it to PROCESSING.
- Exclusive locking prevents multiple operators from processing the same request simultaneously.
- The information needed for Pay-easy payment (biller number, customer number, confirmation number, amount) is displayed in an easy-to-copy format.

5.12 Manual Payment Confirmation

- In the initial PoC, an authorized operator pays **manually** using the Pay-easy function of the corporate internet banking service.
- Operators record the payment result (success/failure), execution date/time, receipt number, etc. in the System.
- On success the status is updated to COMPLETED; on failure to FAILED, with the reason for failure recorded.
- For FAILED requests, operators can choose to reprocess or refund after resolving the cause. (Refund flow details to be determined.)
- In the PoC, Pay-easy payment confirmation itself is simulated.

5.13 Auditing and Reporting

- All payment request status changes, reconciliations, payment executions, and configuration changes are recorded in the audit log.
- Audit logs include the operator, operation, before/after values, and date/time.
- Audit logs cannot be deleted or tampered with.
- Reports such as transaction counts and amounts by period and number of failures can be generated.

5.14 Payment Status Management

Payment requests are managed with the following statuses.

Status	Description	Transitions from
CREATED	Payment request has been created	–
PENDING_TRANSFER	Waiting for customer funds	CREATED
FUNDS_RECEIVED	Incoming funds have been identified and reconciled	PENDING_TRANSFER

Status	Description	Transitions from
READY_FOR_PAYMENT	Payment request has passed validation	FUNDS_RECEIVED
PROCESSING	Domestic bill payment is being processed	READY_FOR_PAYMENT
COMPLETED	Successful payment has been confirmed	PROCESSING
FAILED	Payment processing failed	PROCESSING
CANCELLED	Payment request was cancelled	CREATED / PENDING_TRANSFER

- Undefined status transitions are not allowed.
- Receipt of a transfer (FUNDS_RECEIVED) does not mean the Japanese bill has been paid (COMPLETED).
- Completion of the international transfer and completion of the domestic payment are clearly distinguished in management and display.

6. Non-Functional Requirements

6.1 Security

- Passwords must be at least 8 alphanumeric characters and are stored hashed.
- All communication is encrypted with HTTPS (TLS).
- Secure authentication and authorization are applied to API calls.
- The admin console restricts operations according to role (principle of least privilege). Multi-factor authentication is recommended for administrators.
- Customers' personal and billing information is properly protected and encrypted at rest.
- Session expiration is configured (separately for customers and administrators; values to be determined).
- Audit logs are recorded for all transaction operations to ensure traceability.

6.2 Compliance

Because the System involves cross-border transfers and third-party bill payments, regulatory compliance is essential. The following must be confirmed and addressed before production operation:

- Customer identity verification (KYC)
- Anti-money laundering (AML)
- Canada: FINTRAC compliance and money services business (MSB) requirements
- Japan: Requirements under the Payment Services Act
- Regulations on cross-border collection and payment agency services
- Personal data protection

* Establishing a Japanese corporation or using a corporate bank account does not automatically authorize the business to handle third-party funds. Production deployment requires confirmation of the applicable regulatory requirements and the necessary commercial agreements.

6.3 Reliability and Consistency

- The same payment request must not be executed more than once (duplicate payment prevention), ensured through idempotency keys and exclusive locking.
- Every incoming transfer must be reliably linked to the correct customer and payment request.
- Exceptions such as mismatches and failures must always be detected and remain listed until resolved.
- Accuracy takes priority over real-time responsiveness.

6.4 Performance

- Screen transitions complete within 2 seconds at most.
- Expected number of users and transactions is to be determined. The PoC assumes a small scale (up to a few hundred per month).

6.5 Responsive Design

- Customer screens can be displayed on PC, tablet, and smartphone.
- The admin console is intended mainly for PC.

6.6 Maintainability

- Design documents are properly managed and updated to enable long-term operation.

- External integrations (Wise, bank) are isolated behind interfaces so that simulated and production implementations can be swapped.

6.7 Scalability

- The architecture supports future automation of payment execution through bank integration, automated reconciliation, and payment scheduling.
- Supported bill categories can be added.

6.8 Availability

- Customer screens target 24/7/365 operation. (Payment processing itself is performed during the agency's business hours.)
- Backups are taken so that data — especially transactions and audit logs — is not lost in the event of failure.

7. Supported Environments

○ = Tested × = Not tested

	Windows	Mac	Android	iOS
Google Chrome	○	○	○	○
Microsoft Edge	○	×	×	×
Safari	×	○	×	○

8. Development Roadmap

Phase 1 — Proof of Concept

Objective: Validate the application workflow without processing real customer funds.

- Implement customer authentication
- Develop bill registration functionality
- Create payment request management
- Simulate CAD-to-JPY transfers
- Implement an administrative payment dashboard
- Simulate manual Pay-easy payment confirmation
- Store payment history and audit logs

Phase 2 — Payment Integration

Objective: Investigate and implement supported payment integrations.

- Evaluate Wise Business and API eligibility
- Design transfer reconciliation
- Introduce payment notifications
- Improve bill validation
- Evaluate corporate internet banking operations
- Complete relevant legal and compliance assessments

Phase 3 — Operational Automation

Objective: Reduce manual payment processing.

- Investigate authorized banking integration options
- Automate supported reconciliation processes
- Introduce payment scheduling
- Improve transaction monitoring
- Expand supported bill categories

* Any payment execution automation depends on the capabilities and contractual permissions of the participating financial institution.

9. Key Technical Challenges

Challenge	Description
Cross-border transfer integration	Confirm whether the proposed Wise integration supports the intended customer payment and Japanese corporate account settlement model.
Domestic bill payment execution	Pay-easy supports payments through participating financial institutions, but corporate internet banking does not necessarily provide a programmable payment API.
Payment reconciliation	Each incoming transfer must be reliably matched to the correct customer and payment request.
Duplicate payment prevention	The same payment request must be prevented from being executed multiple times.
Transaction consistency	International transfer completion must be distinguished from domestic bill payment completion.

Challenge	Description
Regulatory compliance	The business model must be evaluated under both Canadian and Japanese financial regulations before handling real customer funds.

10. Budget and Infrastructure

- Budget: To be determined (minimal configuration during the PoC).
- Infrastructure: To be determined. Cloud usage (e.g., AWS) is assumed, with separate development and production environments.
- Payment-related fees (Wise transfer fees, Pay-easy usage fees, bank fees) will be investigated and estimated in Phase 2.

11. Project Goal

The long-term goal is to build a secure, scalable cross-border bill payment platform that makes it easy for residents of Canada to manage their financial obligations in Japan.

The initial focus is on validating the customer experience, the transaction lifecycle, and payment agency operations through a PoC. Future development depends on the availability of suitable banking integrations, payment service partnerships, and regulatory approvals.

12. Disclaimer

- This project is an independent FinTech concept and software development initiative.
- It is not affiliated with, endorsed by, or officially partnered with MyBeacon, Wise, Pay-easy, YES BANK, Bharat Connect, or NTT DATA.
- All external integrations and business relationships described in this document are conceptual unless explicitly identified as implemented and authorized.
- No real customer funds are processed as part of the initial PoC.